Kinetic Energy Calculations
Name:_____

Energy is measured in _____, the symbol is ___.

Kinetic energy = the energy of m_____ is a form of energy so it is also measured in _____.

We can work out how much kinetic energy a moving object has by using the following formula

Kinetic energy= ½ X Objects mass X (objects speed)^2

$$E_k = ½ ms^2$$

Example. A girl has a mass of 60Kg. She is running at a speed of 2m/s. How much kinetic energy does she have?

Kinetic energy= ½ x Objects mass X (objects speed)^2

Kinetic energy= ½ x 60 kg X (2m/s)^2

E_k= ½ x 60 X 4

E_k = ½ x 240

E_k = 120 J

1. A man has a mass of 80Kg and is running at a speed of 2m/s. How much kinetic energy does he have?

Kinetic energy= ½ X Objects mass X (objects speed)^2

Kinetic energy= ½ X _____ X (_____)^2

E_k = ½ X _____ X _____

E_k =½ X _____

E_k = _____ *(don't forget the units)*

2. A car has a mass of 1000kg and is moving at 4m/s. How much kinetic energy does it have?

$$E_k = ½ ms^2$$

Kinetic energy= ½ X _____ X (_____)^2

E_k =

=

=

3. A ball has a mass of 0.5 kg and is moving at 4 m/s. What is its kinetic energy?

$$E_k = ½ ms^2$$

4. Two men are running at 2m/s. One man has a mass of 70kg and the other man has a mass of 80kg. Which man has the greatest kinetic energy?

5. Harry’s trolley contains 2kg of flour and 1kg of apples. Tina’s trolley contains 1kg of pears and 2kg of sugar. They push both their trolleys down the hill with the same force. Which trolley has the most kinetic energy? Why?